

When Recognition Meets Recall

AIAS v0.18 — Indie Fragrance

May 2026 · Pablo Ulpiano González Castro

The dissociation construct generalizes. The Identity-Load moderator is real — but bounded.

Nine cases of Recognition without Recall, on a same-language indie-fragrance substrate. The Iwachu pattern generalizes — it is not a cross-cultural artifact but a measurable mechanism of AI-mediated retrieval. Six of the nine cases concentrate in the highest-Identity-Load cell, where 75% of brands exhibit the pattern. The Identity-Load moderator operates across the three-leg v0.16 / v0.17 / v0.18 history, bounded by substrate-specific qualifications. Three pre-registered hypotheses; three PARTIAL verdicts that say more together than apart.

The AIAS™ Presence Measurement Protocol v1.4 multi-component construct (Recognition × Recall) survives its first generalization test. Nine Iwachu-pattern cases on a same-language indie-fragrance substrate, concentrated in the highest-Identity-Load cell (6 of 8 brands, 75%). Three pre-registered hypotheses; three PARTIAL verdicts. Read together, they describe a moderator that operates through layer-specific channels and a methodology layer measurably closer to canonical.

v0.18 was designed to resolve two questions on a single substrate: the Identity-Load moderator left at AMBIGUOUS by the v0.16/v0.17 panel (v0.16 PARTIAL, v0.17 FALSIFIED on panel inadequacy), and the generalization of the Recognition × Recall dissociation pattern first identified on the v0.17 Japanese-cell substrate.

The substrate is indie fragrance, sampled across three cells stratified by Identity Load: mass-prestige (medium IL), designer-niche (medium-high IL), and indie/artisan (high IL). 24 brands worldwide pre-floor, 8 per cell, English-language anchored throughout. Phase B uses three category-anchored query frames: niche, independent, and perfumistas. The locked six-slot reference panel from v0.17 is carried forward.

All three pre-registered hypotheses returned PARTIAL verdicts. The Recognition × Recall dissociation generalizes across cells (9 cases in 2 of 3 cells) but with cell-clustering in the highest-IL cell (75% of Cell B brands exhibit the pattern). The Identity-Load moderator operates across the three-leg history with substrate-specific qualifications. The

H_Regime4_indie_fragrance verdict resolves at C3 ranking coherence: C1 panel adequacy and both C2 conditions clear, but only one of three cells satisfies the per-cell 📈 0.50 floor.

For senior brand leaders managing identity-bearing categories: the gap between what an AI knows about your brand (Recognition) and what it will say about your brand (Recall) is a measurable, category-level quantity. v0.18 puts a number on the gap — and identifies the substrate conditions under which it widens.

What we measured

Two acquisition phases scored by pre-registered decision rules. Phase A measures **Recognition** — whether the AI panel recognizes a brand as belonging to the category. Phase B measures **Recall** — whether the AI panel surfaces the brand when asked to enumerate category members. Phase D computes within-cell rank correlation between the two.

Phase A asks each of the six reference panel models a single question per brand: *'Is this brand commonly recognized as a niche fragrance?'* The brand's Recognition score (C_P, range 0..6) is the count of recognition-positive responses across the panel. Per-cell pivot anchoring follows a cascade: the first brand achieving C_P = 6/6 becomes the cell pivot; remaining cell brands are probed for the dissociation analysis.

Phase B asks the same six models three category-anchored queries — *'What are the best niche fragrances?'*, *'Recommend high-quality independent fragrance brands.'*, and *'What fragrances do industry insiders and perfumistas recommend?'* — and scans each response for brand mentions across the 24-brand registry. Each brand has at most 18 observation slots (3 queries × 6 models).

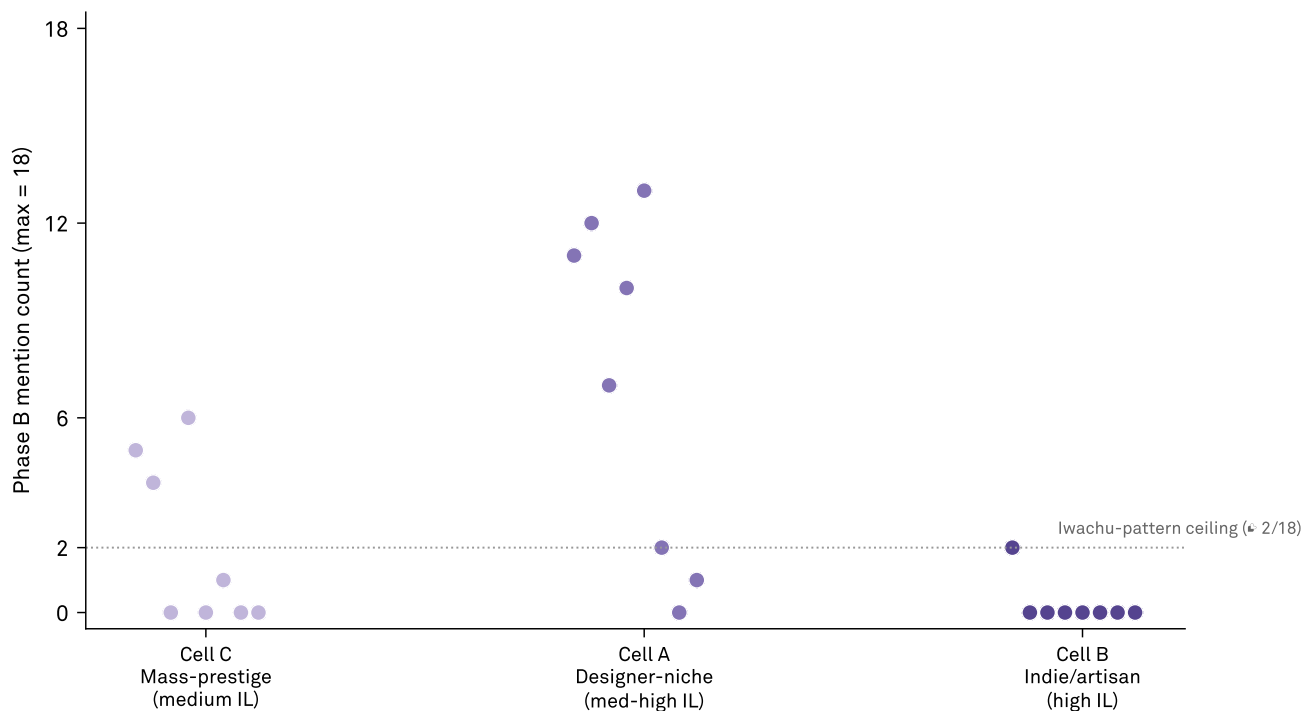
The Recognition × Recall dissociation threshold (the Iwachu-pattern case, novel for v0.18 testing) requires C_P ≥ 5/6 AND mention rate ≥ 2/18 — the brand is recognized in 5+ of 6 single-brand probes, but surfaces in 2 or fewer of 18 enumeration responses. The verdict cascade for H_Regime4_indie_fragrance runs C1 (panel adequacy) → C2 within-cell → C2 IL-gradient → C3 ranking coherence per pre-reg r4 §4.0. The pre-registration is locked at commit 183386c with C2/C3 numerical thresholds locked at commit 1195cb8.

FINDING 01

Recognition meets Recall: the IL-gradient at first glance

Phase B mention rate distribution per cell

v0.18 indie fragrance — 24 brands × 3 cells, IL-gradient ascending C → A → B



Source: AIAS™ Presence Measurement Protocol v1.4 (SSRN 6799479) · v0.18-prereg-r1 · osf.io/ec6wh/v18/

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Figure 1 · Phase B mention-rate distribution per cell. Box-and-whisker plots show within-cell distribution of brand mention counts across the 3 (category queries) × 6 (reference LLMs) = 18 measurement cells. Cell A (designer-niche) shows distributed Recall (Le Labo 17/18, MFK 16/18, Diptyque 15/18 lead a broad coverage). Cell B (indie/artisan) shows near-total Recall attrition (only D.S. & Durga with non-zero mentions, at 2/18). Cell C (mass-prestige) surfaces sporadically (Tom Ford 4, Chanel 3, Versace and Dior 1 each) despite failing the niche-fragrance Recognition probe — evidence of a parallel cultural-footprint Recall channel.

Phase A Recognition shows a sharp Identity-Load gradient. Cell A (designer-niche) anchored Maison Francis Kurkdjian at C_P = 6/6 on the first cascade step; 7 of 8 cell brands scored 6/6 (mean C_P ≈ 5.88/6, near-saturation). Cell B (indie/artisan) anchored D.S. & Durga also at first cascade step, with mean C_P ≈ 5.00/6 across the cell (range 3 to 6). Cell C (mass-prestige) exhausted its full cascade without anchoring a pivot — no Cell C brand achieved C_P = 6/6 (mean

C_P ≈ 0.25/6; 7 of 8 brands at 0/6, Tom Ford alone at 2/6).

The Cell C → Cell A transition is a step function, not a smooth gradient. The 'niche fragrance' Recognition probe maps cleanly onto designer-niche and indie/artisan houses (Cells A and B) but cleanly off mass-prestige designer fragrance (Cell C). The asymmetry isn't a panel-construction error; it's the IL-gradient operating on Recognition itself. Mass-prestige fragrance is, by definition, not niche.

The Cell A > Cell B inversion (5.88 vs. 5.00) is also substantive. The lexical anchor 'niche fragrance' maps most tightly to canonically-labeled designer-niche houses (Le Labo, Frederic Malle, Diptyque) than to the newer American-DTC indie houses (Henry Rose, Phlur, Ellis Brooklyn) that occupy an adjacent vocabulary space — 'indie', 'clean', or 'artisan' rather than 'niche' specifically. The Recognition gradient is real but operates at the threshold-and-vocabulary level, not as a smooth monotone function of Identity Load.

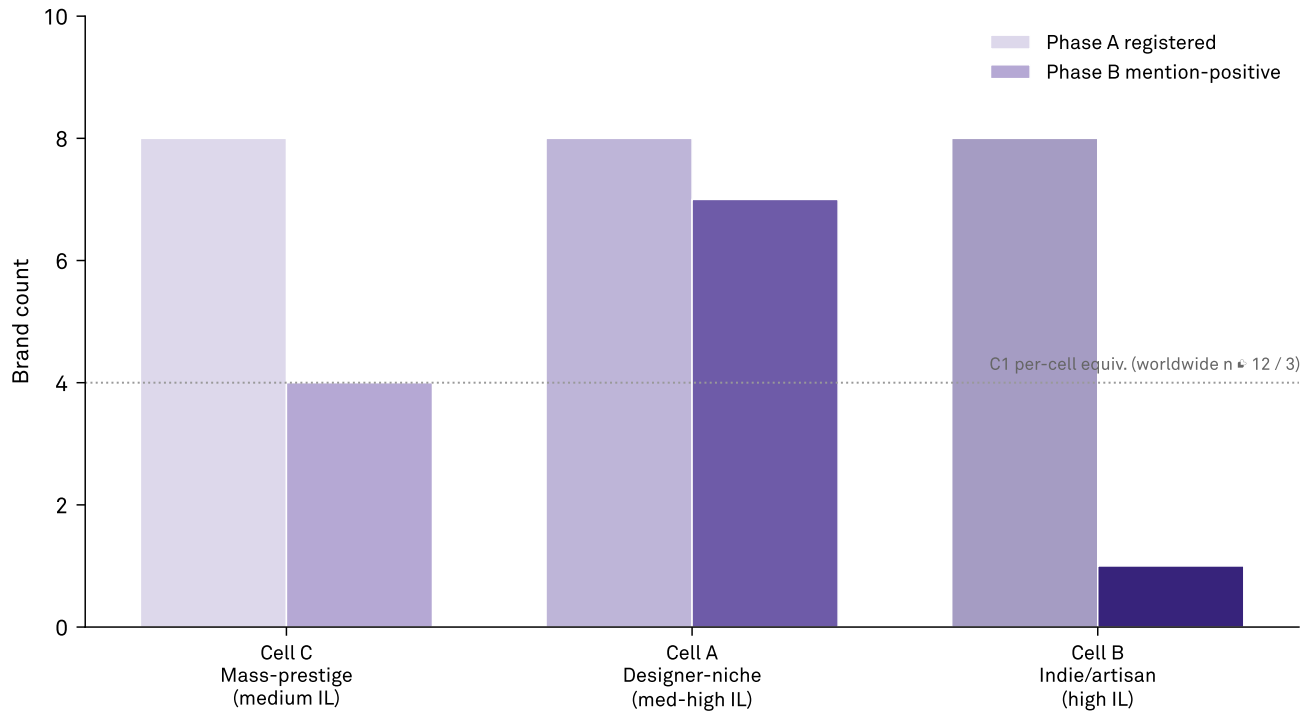
Phase B Recall shows a parallel but differently-shaped gradient. Cell A's mentions distribute across many recognized houses (Le Labo 17/18, MFK 16/18, Diptyque 15/18, Frederic Malle 12/18, Byredo 10/18 — broad coverage, top-2 share 0.446 below the C2 within-cell threshold of 0.50 because Recall is too rich, not too sparse). Cell B shows near-total Recall attrition — 7 of 8 cell brands receive zero mentions across 18 observations; only D.S. & Durga surfaces, and only at 2/18. Cell C surfaces sporadically through general fragrance-discourse channels (Tom Ford 4, Chanel 3, Versace 1, Dior 1) despite failing the niche Recognition probe.

FINDING 02

Cell attrition: panel registered to Phase B Recall-positive

Cell attrition: Phase A registered → Phase B mention-positive

Light bars: Phase A panel (n=8/cell). Dark bars: brands with ≥ 1 mention across 18 Phase B observations.



Source: AIAS™ Presence Measurement Protocol v1.4 (SSRN 6799479) · v0.18-prereg-r1 · osf.io/ec6wh/v18/

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Figure 2 · Cell attrition from Phase A registered panel to Phase B mention-positive set. Light bars are the registered panel (n = 8 per cell); dark bars are brands with ≥ 1 mention across the 18 Phase B observations. Cell A retains 7 of 8 (only Memo Paris is mention-zero); Cell B retains 1 of 8 (D.S. & Durga alone); Cell C retains 4 of 8 (the four with cultural-footprint discourse channels). The cell-attrition pattern is non-monotonic across IL tiers — the IL-gradient operates as dissociation pattern at the high-IL tier rather than as Pareto concentration.

Brand survival from the pre-registered Phase A panel (24 brands; 8 per cell) to the Phase B mention-positive set (12 brands; 1 + 4 + 7 per cell) traces the IL-gradient at the operational layer. Cell A retains 7 of 8 brands (only Memo Paris is mention-zero); Cell C retains 4 of 8 (the four with cultural-footprint discourse: Tom Ford, Chanel, Versace, Dior); Cell B retains 1 of 8 (D.S. & Durga alone surfaces, at 2 mentions of 18). The cell-attrition pattern is the IL-gradient at the Recall layer — and it is non-monotonic across cells.

Cell B's near-total attrition is the methodologically significant outcome. The cell's brands have strong Recognition (mean C_P = 5.00, range 3-6, with five brands at C_P ≥ 5/6) but the panel of LLMs does not surface them when asked to enumerate the category. This isn't 'the AI doesn't know these brands' — it's 'the AI knows these brands and does not retrieve them.' The dissociation operates at scale: most of Cell B exhibits the pattern, not just a single anchor brand.

Cell C's surprising 4 of 8 retention — despite zero Recognition for 'niche fragrance' — points to a parallel Recall channel that operates on cultural-footprint terms independent of category-anchored Recognition. AI systems will include Chanel in a niche-fragrance list while not actually classifying Chanel as niche. This sub-threshold channel is registered for follow-up in v0.19+ as a sensitivity finding; it does not change the v0.18 verdicts but it does suggest the multi-component construct may benefit from a third Recall mode (cultural-footprint vs. category-anchored).

The cell-attribution pattern fed directly into the C3 ranking-coherence verdict. Phase D measures within-cell rank correlation between Recognition (C_P) and Recall (mention count); when one dimension has minimal variance, degenerates. Cell A's flat Recognition (7 of 8 at C_P = 6/6) collapses the rank distinction on the x-axis; Cell B's flat Recall (7 of 8 at zero mentions) collapses it on the y-axis. Only Cell C exhibits sufficient variance in both dimensions to produce a meaningful ($\rho = 0.615$; the only cell clearing the 0.50 floor).

FINDING 03

Nine cases of dissociation, concentrated in the highest-IL cell

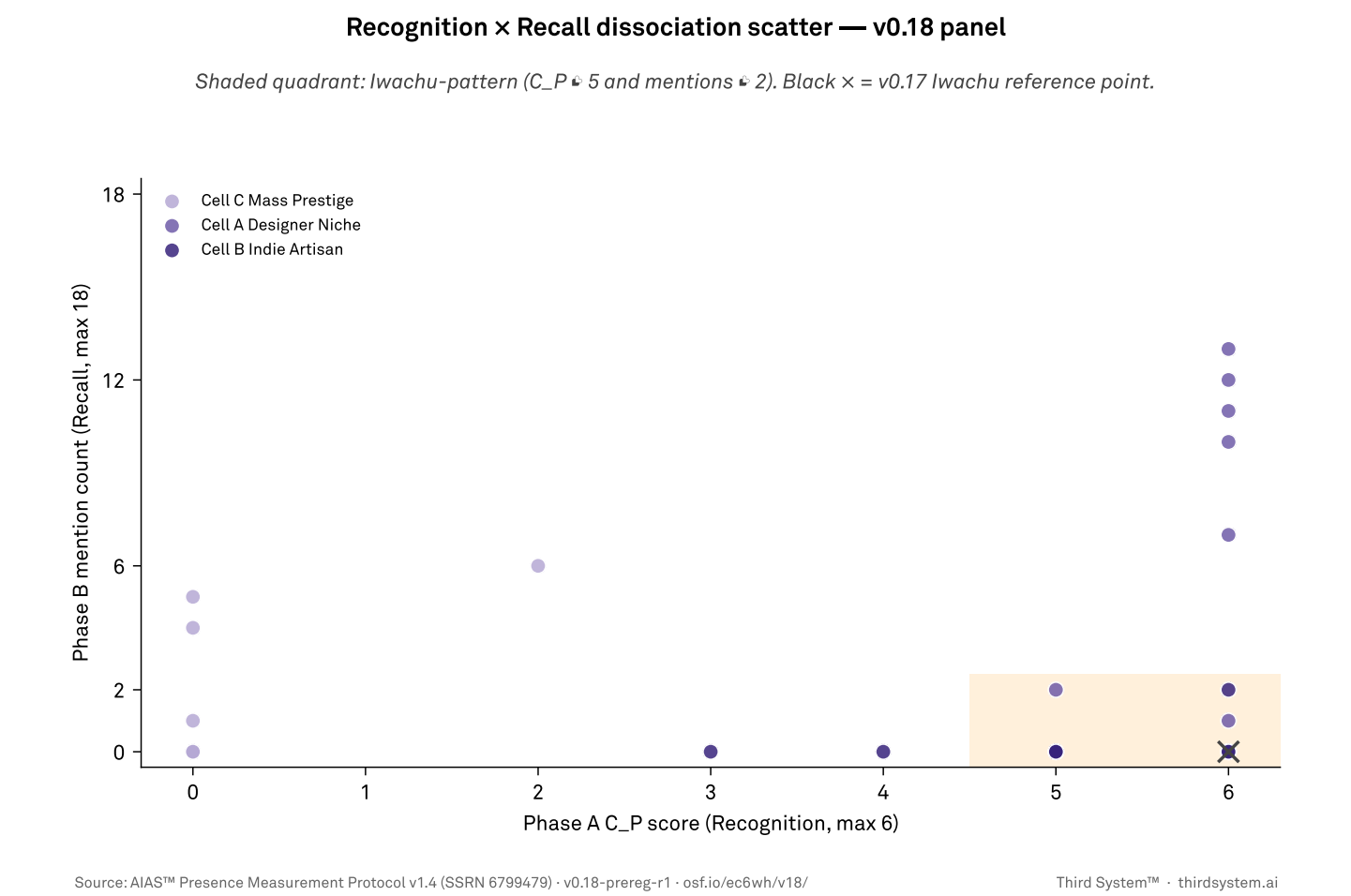


Figure 3 · Recognition x Recall dissociation scatter for the v0.18 panel. Phase A C_P anchoring score (x-axis, 0–6 across the reference panel) versus Phase B mention count (y-axis, 0–18 across the query battery). Shaded quadrant: lwachu-pattern (C_P ⚡ 5 and mentions ⚡ 2). Black x marks the v0.17 lwachu reference anchor (C_P = 6, mentions = 0). v0.18 contributes 9 cases to the shaded quadrant: 3 in Cell A (Comme des Garçons Parfums, Memo Paris, Etat Libre d'Orange) and 6 in Cell B (D.S. & Durga, Boy Smells, Heretic Parfum, Vyrao, Phlur, Snif). Cell C contributes zero by Recognition-floor design — no Cell C brand satisfies C_P ⚡ 5. The construct generalizes from cross-cultural anchor to same-language substrate.

Nine lwachu-pattern cases (Phase A C_P ⚡ 5/6 AND Phase B mentions ⚡ 2/18) on a same-language indie-fragrance substrate. Before v0.18, the v1.4 multi-component construct rested on a single empirical anchor — the lwachu dissociation observed on the v0.17 Japanese-cell substrate. A single case is insufficient ground for a canonical construct claim, and the cross-cultural framing left open the possibility that the

dissociation was a Western-language LLM training-data bias artifact rather than a substantive mechanism. v0.18 closes that question.

Cell A contributes three cases: Comme des Garçons Parfums (C_P = 5/6, mentions = 2/18), Memo Paris (6/6, 0/18), and Etat Libre d'Orange (6/6, 1/18). Cell B contributes six cases: D.S. &

Durga (6/6, 2/18), Boy Smells (5/6, 0/18), Heretic Parfum (6/6, 0/18), Vyrao (6/6, 0/18), Phlur (5/6, 0/18), and Snif (5/6, 0/18). Cell C contributes zero cases by design — the threshold requires $C_P \geq 5/6$, which no Cell C brand achieves (maximum is Tom Ford at 2/6).

The cell-concentration pattern is the methodological headline. If the dissociation construct were noise-driven or artifact-driven, the cases would distribute roughly uniformly across cells. Instead they concentrate where category Identity Load is highest — 75% of Cell B brands exhibit the pattern. This is consistent with the moderator hypothesis and with the substantive theory that AI mediation amplifies retrieval inequality in identity-bearing categories. The v1.4 construct

moves from 'anchored by one cross-cultural case' to 'anchored by ten cases across two substrate families, with a substantive IL-pattern in the second substrate.'

The verdict routes to **DISSOCIATION_PARTIAL** rather than DISSOCIATION_GENERALIZED because Cell C contributes zero cases. This is the asymmetric design of the construct — it captures Recognition-without-Recall, not Non-Recognition-with-Non-Recall — and the Cell C zero is therefore a feature of the design, not a failure of generalization. For AIAS™ 1.0's canonical methodology layer, v0.18 supplies the empirical anchor v1.4 requires before the multi-component Recognition × Recall axis can ship as canonical.

Limitations

Single-substrate constraint on the generalization claim.

v0.18 generalizes the multi-component construct from one substrate (v0.17 cross-cultural anchor) to two (v0.17 + v0.18 same-language). Multi-substrate generalization across the consumer-category spectrum remains future work. v0.19+ should probe substrates with shifted IL profiles and category-vocabulary structures.

LLM panel temporal validity. The six-slot reference panel (claude-opus-4-5, claude-sonnet-4-5, gpt-4o, gpt-4o-mini, gemini-2.5-flash, gemini-2.5-flash-lite) reflects model versions in market as of mid-2026. Provider model substitutions over future-phase horizons are expected and documented in the DEVIATIONS protocol. v0.18 dissociation cases should be read as the pattern observed against the locked panel at acquisition, not as permanent substrate properties.

C2 IL-gradient guard's degenerate case for Cell B. Cell B's top-2 share = 1.000 emerges from Recall sparseness (D.S. & Durga alone, with 2 mentions of 18) rather than from a healthy Pareto distribution. The IL-gradient separation guard clears against a near-empty Cell B baseline. The substantive interpretation rests on the §05 dissociation pattern, not on the C2 mention concentration metric alone.

C3 ranking coherence as a within-cell variance test. Phase D requires within-cell variance in both Recognition and Recall. Cell A's flat Recognition (7 of 8 brands at C_P = 6/6) and Cell B's flat Recall (7 of 8 brands at zero mentions) both degenerate the rank-correlation signal. Future operationalization may benefit from supplementary measures that do not require simultaneous variance in both dimensions.

What's next

Three pre-registerable directions emerge cleanly from the v0.18 verdicts. None requires retracting v0.18; each addresses a specific v0.18 finding.

Substrates with greater within-cell Recognition variance.

The C3 failure mode in v0.18 — Cell A's near-saturation and Cell B's near-zero Recall — both degenerate the within-cell rank correlation. A substrate where Recognition and Recall both vary meaningfully within each cell would test C3 cleanly without confounding by variance collapse.

Finer Identity-Load stratification within the high-IL tier.

v0.18's high-IL cell (indie/artisan) carries 75% of dissociation cases. A substrate with finer IL stratification within the high-IL tier would test whether the dissociation × IL relationship operates monotonically within the tier, or whether it is bounded by an additional threshold structure.

Cell C internal variance — icon-tier vs. mainstream-tier.

Tom Ford's marginal Recognition ($C_P = 2/6$ vs. 0 for the rest of Cell C) and elevated Recall (4 mentions) suggest a sub-threshold Recognition variance operating within the mass-prestige tier. v0.19+ could profitably stratify mass-prestige into icon-tier (Chanel, Dior, Hermès) and mainstream-tier (Marc Jacobs, Calvin Klein) sub-cells to test whether the cultural-footprint Recall channel operates uniformly or stratifies further.

Each direction is independently preregisterable and leaves v0.18's verdicts unchanged. The pre-registration discipline across r1–r4 of v0.18 — locking panel, hypotheses, thresholds, and verdict matrices ex-ante before any acquisition; documenting Cell C's cascade exhaustion as a substantive finding rather than retroactively substituting alternates — is itself the protocol-level commitment that distinguishes the AIAS™ program. v0.19 will inherit the same discipline.

The three verdicts

Three orthogonal pre-registered hypotheses; three PARTIAL verdicts. Read together, they describe a moderator that operates through layer-specific channels and a multi-component construct that generalizes with cell-level concentration in the highest-IL tier.

H	Pre-registered prediction	Result	Status
H_Regime 4	C1 (n = 12) → C2 within-cell (top-2 = 0.50) → C2 IL-gradient (B – C = 0.10) → C3 (per-cell = 0.50 in 2 of 3 cells)	C1 (n = 24); C2 within-cell (Cells B, C); C2 IL-gradient (0.3125); C3 (only Cell C clears, = 0.615)	PARTIAL
H_IL_mod	Joint moderator confirmation across v0.16, v0.17, v0.18 legs	v0.16 PARTIAL × v0.17 FALSIFIED × v0.18 PARTIAL → joint PARTIAL (moderator operates with substrate-specific qualifications)	PARTIAL
H_dissoc	Iwachu pattern generalizes to same-language substrate (1 case per cell GENERALIZED; cell-clustered PARTIAL)	9 cases in 2 of 3 cells (Cell A: 3; Cell B: 6; Cell C: 0 by Recognition-floor design)	DISSOCIATION_PARTIAL

Hypothesis details

The three verdicts in more detail, with the resolution path each took through the pre-registered decision cascade.

Resolved at C3 (ranking coherence). C1 panel adequacy clears cleanly (worldwide n = 24, well above the floor of 12). C2 within-cell clears: Cell B (top-2 = 1.000) and Cell C (top-2 = 0.688) both meet the 0.50 threshold. C2 IL-gradient separation clears: Cell B 1.000 – Cell C 0.688 = 0.3125, above the 0.10 floor. C3 ranking coherence fails: only Cell C clears the per-cell = 0.50 threshold (= 0.615); Cells A and B fail (= 0.247 and 0.434 respectively). 1 of 3 cells clearing is below the 2-of-3 requirement. The failure mode is informative — Cell A's flat Recognition and Cell B's flat Recall both degenerate the within-cell rank correlation.

Three-leg joint verdict over v0.16 (PARTIAL on kitchen knives, SSRN 6791999), v0.17 (FALSIFIED on premium kitchenware, SSRN 6802261, with the falsification grounded in panel inadequacy rather than substantive moderator failure), and v0.18 PARTIAL (this report). The joint matrix routes to PARTIAL: the moderator operates with substrate-specific qualifications.

The v0.18 leg exposes that the moderator acts through layer-specific channels — as a Recognition-floor effect at the medium-IL / medium-high-IL boundary, and as a dissociation pattern at the high-IL tier. The moderator is not a single-mechanism construct.

Nine Iwachu-pattern cases identified across 2 of 3 cells (Cells A and B). The §5.2 verdict matrix routes 'cases present but cell-clustered (2 of 3)' to DISSOCIATION_PARTIAL. The v1.4 multi-component construct generalizes from the cross-cultural Japanese-cell anchor (v0.17 Iwachu) to a same-language IL-gradient substrate. Cell C contributes zero cases by construction — the Iwachu-pattern threshold requires C_P = 5/6, which no Cell C brand achieves. This is the asymmetric design of the construct (Recognition-without-Recall, not Non-Recognition-with-Non-Recall), and the Cell C zero is a feature, not a failure.

About the Third System

The Third System describes a shift in how consumers make decisions: instead of browsing and comparing options directly, many now rely on AI systems to generate recommendations. In this environment, brands compete not only for human attention but for inclusion within AI-generated outputs. This creates a new competitive layer where visibility is determined by how AI systems interpret and surface brands.

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Methodology

This report presents findings from Third System's AI Availability research program. The AI Availability Score (AIAS) is an early-stage measurement framework introduced in Tri-System Brand Growth (Gonzalez Castro, 2026). Findings should be used directionally. Construct validity — the correlation between AI Availability and consumer behavior — has not yet been established and is the subject of Phase 3 of the research program. Methodology is published in full and versioned at thirdsystem.ai/methodology (current: v1.4). Registry construction is curated and reflects the framework's view of each category's competitive set; revisions are documented in the methodology log.

Citation

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Underlying datasets

Pre-registration and DEVIATIONS log: osf.io/ec6wh/v18/PRE_REGISTRATION_v0_18.md and osf.io/ec6wh/v18/DEVIATIONS.md

Phase A and Phase B acquisition data: osf.io/ec6wh/v18/data/

Scoring code, figures, and verdict outputs: osf.io/ec6wh/v18/scripts/ and osf.io/ec6wh/v18/data/verdicts/

Methodology log: thirdsystem.ai/methodology and osf.io/ec6wh.

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